

Notes: Cohen and Einav (AER, 2007)
Estimating Risk Preferences from Deductible Choice

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1 Big picture

- **Question.** How risk averse are individuals, how heterogeneous are they in risk preferences, and how do risk preferences vary with observable characteristics?
- **Setting.** The paper studies deductible choices in auto insurance contracts sold by a direct insurer in Israel from November 1994 to October 1999.
- **Key data advantage.** The authors observe more than 100,000 individual-specific menus of deductible-premium combinations, the deductible choice, policyholder characteristics, policy duration, and realized claims.
- **Core empirical idea.** Choosing a lower deductible costs a higher premium but reduces the payment in the event of a claim. For a given claim probability, this choice reveals a lower bound on absolute risk aversion.
- **Main complication.** People choosing low deductibles may be more risk averse, more likely to have accidents, or both. The paper therefore models unobserved heterogeneity in both claim risk and risk aversion.
- **Benchmark model.** Claims are generated by an individual-level Poisson process. The individual knows her claim rate. The log claim rate and log coefficient of absolute risk aversion are jointly normally distributed conditional on observables.
- **Main findings.** Risk aversion is highly heterogeneous and skewed. The median individual is close to risk neutral for roughly \$100 bets, while the mean is much more risk averse because of a right tail.
- **Observable heterogeneity.** Women are more risk averse than men; absolute risk aversion is U-shaped in age; and several proxies for income or wealth are positively associated with absolute risk aversion.
- **Risk versus preference heterogeneity.** Unobserved heterogeneity in risk aversion is larger and more important for pricing than unobserved heterogeneity in risk.
- **Economic takeaway.** Insurance choices reveal that people differ substantially in willingness to bear small financial risks. This heterogeneity can materially change optimal insurance pricing and the interpretation of adverse selection.

2 Incremental contribution of the paper

- **From representative risk aversion to heterogeneous risk aversion.** Much earlier empirical work estimates a representative risk-aversion parameter. This paper estimates a distribution of risk preferences across individuals.
- **From hypothetical or lab choices to market choices.** The evidence comes from actual insurance purchases by real consumers, not from hypothetical surveys, game shows, or laboratory lotteries.
- **A clean insurance-margin contribution.** Deductible options differ mainly in financial terms - premiums and deductibles - so the choice is unusually well suited for inferring risk preferences.

- **Joint modeling of risk and preferences.** The paper explicitly separates selection on risk from selection on risk aversion. This is crucial because low deductible buyers have higher claim rates.
- **Demand-side identification.** Instead of relying on optimality of supplied contracts, the paper uses observed choices from individual-specific menus and realized claim data.
- **Multidimensional private information.** The paper shows that insurance markets may be shaped by two unobserved dimensions: expected cost to the insurer and willingness to pay for insurance.
- **Pricing implications.** The counterfactuals show that ignoring risk-aversion heterogeneity can drastically change conclusions about profitability and optimal deductible pricing.
- **Methodological contribution.** The paper provides a structural framework for estimating preference heterogeneity from customized contracts, realized risk outcomes, and discrete choices.

3 Data and measurement

3.1 Economic environment

- The data come from one Israeli auto insurance company that sold directly to customers rather than through agents.
- The sample contains 105,800 new policyholders who bought annual policies from November 1994 to October 1999.
- The insurer was a new direct insurer. Early in the sample, it had a substantial price advantage relative to conventional insurers.
- The paper focuses on new policyholders' first contracts to avoid selection created by later renewal decisions.
- The policy resembles comprehensive auto insurance. It covers many car-damage risks but not bodily injury or third-party injury.

Interpretation: The setting is attractive because the insurer records the exact menu offered to each customer and the later claim realizations. The main external-validity issue is that the sample consists of consumers willing to buy direct insurance.

3.2 Deductible menu

Each individual is offered four deductible-premium combinations: low, regular, high, and very high deductible. The regular premium is a deterministic function of observed characteristics. The regular deductible is:

$$d_i = \min\{0.5p_i, \text{cap}_t\},$$

where the deductible cap varies over time but not across individuals.

The low-deductible option has a higher premium and lower deductible:

- low premium is approximately 1.06 times the regular premium;

- low deductible is approximately 0.6 times the regular deductible;
- high and very high deductibles are rarely chosen.

Interpretation: The key price variation comes from changes in the deductible cap and from a period of experimentation with multipliers. This creates variation in the relative price of extra insurance coverage.

3.3 Choices and claim outcomes

- The paper mainly analyzes the choice between low and regular deductibles.
- About 17.8% of policyholders choose the low deductible and 81.1% choose the regular deductible.
- Only about 1% choose the high or very high deductible options, so the paper treats them as regular-deductible choices in the main analysis.
- The average annual claim rate is about 0.245.
- Claim rates differ by choice: low-deductible buyers have a claim rate of about 0.309, while regular-deductible buyers have a claim rate of about 0.232.

Interpretation: The raw data already show adverse selection: people who buy more insurance have more claims. But this correlation does not say whether the selection is caused by risk, risk aversion, or both.

3.4 Covariates

The data include policyholder demographics, car characteristics, driving variables, young-driver information, company-year indicators, and zip-code-level income or wealth proxies. Important variables include age, gender, marital status, education, immigrant status, car value, car age, engine size, license years, claims history, business use, and whether the policy allows any driver.

Interpretation: The rich covariates help predict claim risk and explain observable variation in risk aversion. However, the paper's central contribution is estimating unobserved heterogeneity after conditioning on these variables.

4 Models and empirical specifications

4.1 Deductible choice under expected utility

Let (p^h, d^h) be the regular high-deductible contract and (p^l, d^l) be the low-deductible contract, with $p^l > p^h$ and $d^l < d^h$. Let λ_i be the annual claim rate and r_i the coefficient of absolute risk aversion. For a short policy interval t , expected utility from contract (p, d) is:

$$v(p, d) = (1 - \lambda t)u(w - pt) + \lambda t u(w - pt - d).$$

The individual is indifferent between the two deductibles when:

$$(p^l - p^h)u'(w) = \lambda \left[u(w - d^l) - u(w - d^h) \right].$$

Using a second-order approximation, define:

$$\Delta p = p^l - p^h, \quad \Delta d = d^h - d^l, \quad d = \frac{d^h + d^l}{2}.$$

Then the approximate indifference condition is:

$$r^*(\lambda) = \frac{\frac{\Delta p}{\lambda \Delta d} - 1}{d}.$$

The low deductible is chosen if:

$$r_i > r_i^*(\lambda_i).$$

Interpretation: Higher risk lowers the required risk aversion needed to choose the low deductible. This is why adverse selection and risk aversion are hard to separate without a structural model.

4.2 Benchmark structural model

The benchmark model assumes:

$$\begin{aligned} \ln \lambda_i &= x_i' \beta + \epsilon_i, \\ \ln r_i &= x_i' \gamma + \nu_i, \end{aligned}$$

with

$$\begin{pmatrix} \epsilon_i \\ \nu_i \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_\lambda^2 & \rho \sigma_\lambda \sigma_r \\ \rho \sigma_\lambda \sigma_r & \sigma_r^2 \end{pmatrix} \right).$$

Claims follow:

$$\text{claims}_i \sim \text{Poisson}(\lambda_i t_i),$$

where t_i is the observed policy duration. The low deductible is chosen when $r_i > r_i^*(\lambda_i)$.

Interpretation: The model uses claim realizations to learn about the distribution of risk and deductible choices to learn about the distribution of risk aversion.

4.3 Estimation

The paper estimates the model using a Gibbs sampler with data augmentation. The latent individual-level variables λ_i and r_i are simulated and then treated as augmented data in the estimation.

- Conditional on the latent risk and risk-aversion draws, the model resembles two linear equations in logs.
- Conditional on risk, log risk aversion follows a truncated normal distribution because the deductible choice imposes a lower or upper bound.
- Conditional on risk aversion, log risk is sampled using claim data and the deductible-choice truncation.

Interpretation: The Bayesian computational approach is mainly a practical way to estimate a model with two latent dimensions and many individuals.

4.4 Reduced-form regressions

The reduced-form analysis estimates:

- a Poisson regression of realized claims on observed characteristics;
- probit regressions of choosing the low deductible on observed characteristics and pricing variables.

Interpretation: These regressions motivate the structural model. Claims and deductible choices are correlated even after observables, indicating selection on unobservables.

4.5 Counterfactual pricing

The paper uses the estimated joint distribution of risk and risk aversion to compute insurer profits under alternative low-deductible designs. Holding the regular contract fixed, the insurer chooses the low deductible and premium trade-off.

Interpretation: The counterfactuals show why estimating the full distribution matters: pricing depends on both demand elasticity from risk aversion and composition effects from adverse selection.

5 Identification and empirical design

5.1 What identifies claim risk

The distribution of risk types is identified primarily from the realized claims and policy durations. A lognormal mixture of Poisson distributions separates persistent heterogeneity in claim rates from random claim realizations.

Interpretation: Claim data are essential because the deductible choice alone cannot tell whether low deductible buyers are high risk or highly risk averse.

5.2 What identifies risk aversion

Conditional on the risk distribution, the deductible choices identify the distribution of risk aversion. If low-risk individuals choose the low deductible, the model attributes this to high risk aversion.

Interpretation: The main information about risk aversion comes from people choosing extra coverage even when their inferred claim risk is not high enough to justify it under risk neutrality.

5.3 Role of menu variation

Variation in the relative price of the low deductible changes the indifference curve in (λ, r) space. These changes help identify the distribution of risk aversion and its correlation with risk.

Interpretation: When different individuals face different price-deductible trade-offs, the same observed choice has different implications for risk aversion.

5.4 Distinguishing risk and risk aversion

The model distinguishes two selection channels:

- **Selection on risk:** high-risk individuals buy more coverage because they expect more claims.
- **Selection on risk aversion:** highly risk-averse individuals buy more coverage even when claim risk is moderate.

Interpretation: The paper finds both channels matter, but unobserved heterogeneity in risk aversion is more important for pricing and profits.

5.5 Key assumptions and caveats

- **No moral hazard.** The benchmark assumes claim rates do not change with deductible choice.
- **Known risk type.** Individuals are assumed to know their own claim rates, though the paper studies incomplete-information robustness checks.
- **Expected utility.** The benchmark uses expected utility and summarizes local curvature by absolute risk aversion.
- **Sample selection.** The sample consists of consumers who bought from a direct insurer, so the estimates may not represent all drivers.
- **Transaction costs.** Accidents may involve costs beyond deductibles, which could affect the interpretation of estimated risk aversion.

Interpretation: Most caveats affect the level or interpretation of estimated risk aversion, but the main qualitative conclusion - large heterogeneity in risk preferences - is robust.

6 Tables and figures

6.1 Table 1: Summary statistics - covariates

Read:

- Table 1 reports the observable characteristics of the 105,800 new customers.
- The mean age is about 41; about 31.6% of policyholders are female.
- The average car value is about 66,958 NIS, with substantial dispersion.
- The table includes demographic variables, car attributes, driving experience, past claims history, young-driver indicators, and company-year indicators.
- These covariates enter both the risk equation and the risk-aversion equation in the benchmark structural model.

Economic message: The table establishes that the insurer observes rich information about policyholders. The remaining variation in choices and claims after these covariates is what identifies unobserved risk and unobserved risk aversion.

6.2 Figure 1: Variation in the deductible cap over time

Read:

- Figure 1 plots the deductible cap used to determine regular deductibles over time.
- The cap changes discretely during the sample period.
- These cap changes create variation in the deductible-premium menu that is not individual-specific.
- The authors use this variation as one source of identification for how deductible choices respond to prices.

Economic message: The figure shows why the paper has meaningful price variation. Individuals with similar observables can face different deductible trade-offs simply because they buy at different times.

TABLE 1—SUMMARY STATISTICS—COVARIATES

Variable		Mean	Std. dev.	Min	Max
Demographics:	Age	41.137	12.37	18.06	89.43
	Female	0.316	0.47	0	1
	Family	Single	0.143	0.35	0
		Married	0.779	0.42	0
		Divorced	0.057	0.23	0
		Widower	0.020	0.14	0
		Refused to say	0.001	0.04	0
	Education	Elementary	0.016	0.12	0
		High school	0.230	0.42	0
		Technical	0.053	0.22	0
		College	0.233	0.42	0
		No response	0.468	0.50	0
Emigrant		0.335	0.47	0	1
Car attributes:	Value (current NIS) ^a	66,958	37,377	4,000	617,000
	Car age	3.952	2.87	0	14
	Commercial car	0.083	0.28	0	1
	Engine size (cc)	1,568	385	700	5,000
Driving:	License years	18.178	10.07	0	63
	Good driver	0.548	0.50	0	1
	Any driver	0.743	0.44	0	1
	Secondary car	0.151	0.36	0	1
	Business use	0.082	0.27	0	1
	Estimated mileage (km) ^b	14,031	5,891	1,000	32,200
	History length	2.847	0.61	0	3
	Claims history	0.060	0.15	0	2
Young driver:	Young	0.192	0.39	0	1
	Gender	Male	0.113	0.32	0
		Female	0.079	0.27	0
	Age	17–19	0.029	0.17	0
		19–21	0.051	0.22	0
		21–24	0.089	0.29	0
		>24	0.022	0.15	0
	Experience	<1	0.042	0.20	0
		1–3	0.071	0.26	0
		>3	0.079	0.27	0
Company year:	First year	0.207	0.41	0	1
	Second year	0.225	0.42	0	1
	Third year	0.194	0.40	0	1
	Fourth year	0.178	0.38	0	1
	Fifth year	0.195	0.40	0	1

Note: The table is based on all 105,800 new customers in the data.

^a The average exchange rate throughout the sample period was approximately \$1 per 3.5 NIS, starting at 1:3 in late 1994 and reaching 1:4 in late 1999.

^b The estimated mileage is not reported by everyone. It is available for only 60,422 new customers.

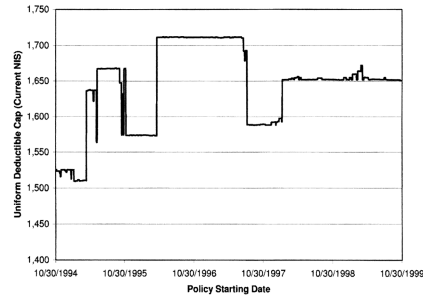


FIGURE 1. VARIATION IN THE DEDUCTIBLE CAP OVER TIME

Notes: This figure presents the variation in the deductible cap over time, which is one of the main sources of pricing variation in the data. We do not observe the cap directly, but it can be calculated from the observed menus. The graph plots the maximal regular deductible offered to anyone who bought insurance from the company over a moving seven-day window. The large jumps in the graph reflect changes in the deductible cap. There are three reasons why the graph is not perfectly smooth. First, in a few holiday periods (e.g., October 1995) there are not enough sales within a seven-day window, so none of those sales hits the cap. This gives rise to temporary jumps downward. Second, the pricing rule applies at the date of the price quote given to the potential customer. Our recorded date is the first date the policy becomes effective. The price quote is held for a period of two to four weeks, so in periods in which the pricing changes, we may still see new policies sold using earlier quotes, made according to a previous pricing regime. Finally, even within periods of constant cap, the maximal deductible varies slightly (variation of less than 0.5 percent). We do not know the source of this variation.

Table 1: Summary statistics - covariates

Figure 1: Variation in the deductible cap

6.3 Table 2A and 2B: Menus, choices, and outcomes

Read:

- Table 2A summarizes the four deductible-premium options and policy outcomes.
- The low deductible averages about 875 NIS, while the regular deductible averages about 1,453 NIS.
- The low premium averages about 3,381 NIS, while the regular premium averages about 3,189 NIS.

- 17.8% choose the low deductible; 81.1% choose the regular deductible; high and very high deductibles are rarely chosen.
- Average annual claim rates are higher for low-deductible buyers than regular-deductible buyers.
- Table 2B shows the same positive relation between claims and low-deductible choice in a cross-tabulation.

Economic message: The table gives the raw adverse-selection fact: people buying more coverage have more claims. The rest of the paper asks how much of that is risk and how much is risk aversion.

6.4 Figure 2: The individual's decision - graphical illustration

Read:

- Figure 2 plots the indifference set between regular and low deductible choices in risk-risk-aversion space.
- The low deductible is optimal for individuals above the curve or to the right of it.
- High-risk individuals need less risk aversion to choose the low deductible.
- The simulated cloud illustrates the joint distribution of risk and risk aversion under the benchmark estimates.

Economic message: The figure makes the identification problem transparent. A low-deductible choice can come from high expected claims, high risk aversion, or both.

Variable		Obs	Mean	Std. dev.	Min	Max
Menu:	Deductible (current NIS) ^a	Low	105,800	875.48	121.01	374.92
		Regular	105,800	1,452.99	197.79	624.86
		High	105,800	2,608.02	352.91	1,124.75
		Very high	105,800	3,763.05	508.53	1,624.64
	Premium (current NIS) ^a	Low	105,800	3,380.57	914.04	13,247.71
		Regular	105,800	3,189.22	862.3	12,497.72
		High	105,800	2,790.57	754.51	10,953.51
		Very high	105,800	2,551.37	689.84	9,997.78
	$\Delta p / \Delta d$		105,800	0.328	0.06	0.3
					0.3	1.8
Realization:	Choice	Low	105,800	0.178	0.38	0
		Regular	105,800	0.811	0.39	0
		High	105,800	0.006	0.08	0
		Very high	105,800	0.005	0.07	0
	Policy termination	Active	105,800	0.150	0.36	0
		Canceled	105,800	0.143	0.35	0
		Expired	105,800	0.707	0.46	0
		Expired	105,800	0.848	0.28	0.005
	Claims	All	105,800	0.208	0.48	0
		Low	18,799	0.280	0.55	0
		Regular	85,840	0.194	0.46	0
		High	654	0.109	0.34	0
	Claims per year ^b	Very high	507	0.107	0.32	0
		All	105,800	0.245	0.66	0
		Low	18,799	0.309	0.66	0
		Regular	85,840	0.232	0.66	0
		High	654	0.128	0.62	0
		Very high	507	0.133	0.50	0

^a The average exchange rate throughout the sample period was approximately \$1 per 3.5 NIS, starting at 1:3 in late 1994 and reaching 1:4 in late 1999.

^b The mean and standard deviation of the claims per year are weighted by the observed policy duration to adjust for variation in the exposure period. These are the maximum likelihood estimates of a simple Poisson model with no covariates.

Claims	Low	Regular	High	Very high	Total	Share
0	11,929 (0.193)	49,281 (0.796)	412 (0.007)	299 (0.005)	61,921 (1.00)	0.8034
1	3,124 (0.239)	9,867 (0.755)	47 (0.004)	35 (0.003)	13,073 (1.00)	0.1696
2	565 (0.308)	1,261 (0.688)	4 (0.002)	2 (0.001)	1,832 (1.00)	0.0238
3	71 (0.314)	154 (0.681)	1 (0.004)	0 (0.000)	226 (1.00)	0.0029
4	6 (0.353)	11 (0.647)	0 (0.000)	0 (0.000)	17 (1.00)	0.0002
5	1 (0.500)	1 (0.500)	0 (0.000)	0 (0.000)	2 (1.00)	0.0003

Notes: The table presents tabulation of choices and number of claims. For comparability, the figures are computed only for individuals whose policies lasted at least 0.9 years (about 73 percent of the data). The bottom rows of Table 2A provide descriptive figures for the full dataset. The numbers in parentheses in each cell represent percentages within each row. The right-hand-side column presents the marginal distribution of the number of claims.

Table 2A/2B: Menus, choices, and outcomes

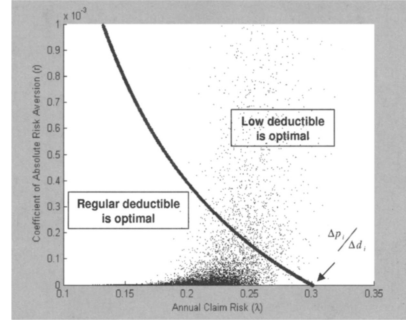


FIGURE 2. THE INDIVIDUAL'S DECISION—A GRAPHICAL ILLUSTRATION

Notes: This graph illustrates the individual's decision problem. The solid line presents the indifference set—equation (7)—applied for the menu faced by the average individual in the sample. Individuals are represented by points in the two-dimensional space above. In particular, the scattered points are 10,000 draws from the joint distribution of risk and risk aversion for the average individual (on observables) in the data, based on the point estimates of the benchmark model (Table 4). If an individual is either to the right of the line (high risk) or above the line (high risk aversion), the low deductible is optimal. Adverse selection is captured by the fact that the line is downward sloping: higher-risk individuals require lower levels of risk aversion to choose the low deductible. Thus, in the absence of correlation between risk and risk aversion, higher-risk individuals are more likely to choose higher levels of insurance. An individual with $\lambda_i > (\Delta p / \Delta d)_i$ will choose a lower deductible even if he is risk neutral, i.e., with probability one (we do not allow individuals to be risk loving). This does not create an estimation problem because λ_i is not observed, only a posterior distribution for it. Any such distribution will have a positive weight on values of λ_i that are below $(\Delta p / \Delta d)_i$. Second, the indifference set is a function of the menu and, in particular, of $(\Delta p / \Delta d)_i$ and d_i . An increase in $(\Delta p / \Delta d)_i$ will shift the set up and to the right, and an increase in d_i will shift the set down and to the left. Therefore, exogenous shifts of the menus that make both arguments change in the same direction can make the sets "cross," thereby allowing us to identify the correlation between risk and risk aversion nonparametrically. With positive correlation (shown in the figure by the "right-bending" shape of the simulated draws), the marginal individuals are relatively high risk, therefore creating a stronger incentive for the insurer to raise the price of the low deductible.

Figure 2: Deductible choice in risk-risk-aversion space

6.5 Table 3: No heterogeneity in risk

Read:

- Table 3 reports reduced-form Poisson and probit regressions.
- The Poisson regression predicts claim counts from observables.
- The probit regressions predict whether the policyholder chooses the low deductible.
- Many variables predict both claims and deductible choice, but not always in the same direction.
- The restricted model without unobserved risk heterogeneity is rejected because it implies implausible risk-aversion estimates.

Economic message: The table motivates the full structural model. Observables alone cannot explain both claims and choices, so the paper needs latent risk and latent risk aversion.

6.6 Table 4: The benchmark model

Read:

- Table 4 reports the main structural estimates for the log claim-rate equation and log risk-aversion equation.
- Women are estimated to have about 20% higher absolute risk aversion than men.
- Risk aversion is U-shaped in age, declining initially and rising after roughly age 48.
- Car value is positively associated with both claim risk and risk aversion.
- The estimated latent correlation between unobserved risk and unobserved risk aversion is positive and large.
- The unconditional mean risk-aversion parameter is far above the median, implying a highly skewed distribution.

Economic message: Table 4 is the central result. It shows that risk preferences vary systematically with observables and that unobserved risk-aversion heterogeneity is large relative to risk heterogeneity.

TABLE 3—NO HETEROGENEITY IN RISK

Variable	Poisson regressions ^a			Probit regressions ^b		
	Dep. var: Number of claims			Dep. var: 1 if low deductible		
	(1)	(2)	(3)	(4)	(5)	(6)
Demographics:	Constant	Coeff	Std. Err.	Coeff	Std. Err.	dP/dX
	Age	-4.3535*	0.7444	-22.4994*	2.5952	-
	Age ²	-0.0074	0.0052	0.9926	0.0464	-0.0034*
	Female	$-9.89 \cdot 10^{-5}$	$5.36 \cdot 10^{-5}$	1.0001	0.001*	$3.39 \cdot 10^{-5}$
	Family	-0.0456*	0.0161	0.9555	0.8003*	0.0165*
	Married	omitted	omitted	omitted	omitted	omitted
	Divorced	-0.1115*	0.0217	0.8945	0.1591	0.0128*
	Widower	0.0717*	0.0346	1.0743	-0.3191	-0.0064
	Other (NA)	0.0155	0.0527	1.0156	-0.3441	0.4573
	Elementary	0.1347	0.1755	1.1442	-2.9518	1.9636
Education	High school	-0.0522	0.0550	0.9491	0.2927	0.4278
	Technical	omitted	omitted	omitted	omitted	omitted
	College	0.0373	0.0388	1.0380	0.5037*	0.2520
	Other (NA)	-0.0745*	0.0197	0.9282	0.5279*	0.1546
	Emigrant	0.0116	0.0184	1.0116	0.0630	0.1540
	Log(Value)	0.0210	0.0163	1.0213	-0.0497	0.1328
	Car age	0.1227*	0.0281	1.1306	1.3283*	0.2342
	Commercial car	0.0187*	0.0042	1.0189	-0.1603*	0.0357
	Log(longevity size)	-0.1394*	0.0336	0.8699	-0.9038*	0.2781
	License years ²	0.2972*	0.0409	1.3461	-0.6924	0.3952
Driving:	Good driver	-0.0204*	0.0024	0.9798	0.1043*	0.0212
	Any driver	0.0002*	$6.77 \cdot 10^{-5}$	1.0002	-0.0015*	0.0005
	Secondary car	-0.0076	0.0191	0.9825	-0.7207*	0.1574
	Business use	0.0564*	0.0169	1.0580	1.0923*	0.1521
	History length	-0.0859*	0.0209	0.9177	0.0038	0.1626
	Claims history	0.1852*	0.0293	1.2034	-0.7951*	0.2737
	Young driver	0.2972*	0.0110	1.0486	1.1218*	0.1450
	Gender	0.6577*	0.0390	1.9304	-0.4654	0.5460
	Male	0.5235*	0.0361	1.6879	-2.8847*	0.6305
	Female	omitted	omitted	omitted	omitted	omitted
Young driver:	Age	-0.1475*	0.0288	0.8629	1.7959*	0.3135
	17-19	omitted	omitted	omitted	omitted	omitted
	20-24	0.0701	0.0532	1.0726	-0.7800	0.7509
	25-34	-0.0267	0.0574	0.9737	-0.5746	0.7732
	>34	-0.2082*	0.0567	0.812	1.7869*	0.7328
	Experience	<1	omitted	omitted	omitted	omitted
	1-3	-0.2416*	0.0458	0.7854	1.0175	0.6577
	>3	-0.2827*	0.0532	0.7538	3.2513*	0.7386
	Company year:	omitted	omitted	omitted	omitted	omitted
	First year	-0.0888*	0.0198	0.9150	-4.4513*	0.1506
Company year:	Second year	-0.0609*	0.0222	0.9334	-5.5808*	0.1620
	Third year	-0.1817*	0.0232	0.8339	-11.8277*	0.2102
	Fourth year	-0.5431*	0.028	0.5810	-14.3206*	0.2778
	Fifth year	omitted	omitted	omitted	omitted	omitted
	σ	105.800	94.000	10.586	94.000	105.800
	Obs.	0.0162	0.1364	0.1296	0.1296	0.1296
	Pseudo R ²	-57.745.9	-	-36.959.5	-	-43.086.9
	Log likelihood	-	-	-	-	-

* Significant at the 5 percent confidence level.

^a BRR = Incidence rate ratio. Each figure should be interpreted as the increase/decrease in claim probability as a result of an increase of one unit in the right-hand-side variable.

^b There are two separate probit regressions reported in this table. Column 2 relies on the deductible choice model and the lognormality assumption. As discussed in the text, by including an additional regressor, $\ln(\text{Lay}(A_1/A_2) - 1)$ (with A_1 predicted from column 1 above), and normalizing its coefficient to -1, we obtain a structural interpretation of this regression, with $\ln(\tau)$ as the dependent variable. Thus, the reported coefficients are comparable to those estimated for the benchmark model. One should be cautious, however, in interpreting these coefficients. Unlike the benchmark model, this regression does not allow unobserved heterogeneity in risk and suffers from some selection bias because observations with high predicted risk rate are omitted (which is why the number of observations is 94,000 rather than the full sample of 105,800). For comparison, column 3 reports the marginal effects from a comparable probit regression that uses the full sample and does not control for pricing and predicted risk through the additional structural regressor. Column 3 does not have a structural interpretation, and its (unreported) coefficients cannot be compared to those estimated from the benchmark model.

TABLE 4—THE BENCHMARK MODEL

Variable	Logit equation		Additional quantities	
	Logit equation	Logit equation	Var-cov matrix (Σ)	
Demographics:	Constant	-1.5406 (0.0073)*	-11.8118 (0.1032)*	
	Age	-0.0001 (0.0026)	-0.0623* (0.0213)	σ_x
	Age ²	$6.24 \cdot 10^{-6}$ ($2.63 \cdot 10^{-5}$)	$6.44 \cdot 10^{-4}$ ($2.11 \cdot 10^{-4}$)	σ_y
	Female	0.0006 (0.0086)	0.2049 (0.0843)*	ρ
	Family	omitted	omitted	
	Single	omitted	omitted	
	Married	-0.0198 (0.0115)	0.1927 (0.0974)*	Unconditional statistics ^a
	Divorced	0.0396 (0.0155)*	-0.1754 (0.1495)	Mean A
	Widower	0.0135 (0.0281)	-0.1320 (0.2288)	Median A
	Other (NA)	-0.0557 (0.0668)	-0.4599 (0.7397)	Std. Dev. A
Education	Elementary	-0.0194 (0.0333)	0.1283 (0.2156)	Mean τ
	High school	omitted	omitted	Median τ
	Technical	-0.0007 (0.0189)	0.2306 (0.1341)	Std. Dev. τ
	College	-0.0277 (0.0124)*	0.2177 (0.0840)*	Corr. (A)
	Other (NA)	-0.0029 (0.0307)	0.0328 (0.0919)	
	Emigrant	0.0000 (0.0090)	0.0001 (0.0651)	Obs.
	Log(Value)	0.0794 (0.0177)*	0.7244 (0.1272)*	
	Car age	0.0053 (0.0023)*	-0.0411 (0.0176)*	
	Commercial car	-0.0719 (0.0187)*	-0.0313 (0.1229)	
	Log(longevity size)	0.1299 (0.0235)*	-0.3195 (0.1847)	
Driving:	License years	-0.0015 (0.0017)	0.0157 (0.0137)	
	License years ²	$-1.83 \cdot 10^{-5}$ ($3.51 \cdot 10^{-5}$)	$-1.48 \cdot 10^{-4}$ ($2.54 \cdot 10^{-4}$)	
	Good driver	-0.0635 (0.012)*	-0.0517 (0.0822)	
	Any driver	0.0360 (0.0105)*	0.3900 (0.0722)*	
	Secondary car	-0.0415 (0.0141)*	0.1209 (0.0875)	
	Business use	0.0614 (0.0134)*	-0.3790 (0.1124)*	
	History length	0.0012 (0.0052)	0.3092 (0.0518)*	
	Claims history	0.1295 (0.0154)*	0.0459 (0.1670)	
	Young driver	0.0525 (0.0253)*	-0.2499 (0.2290)	
	Gender	omitted	omitted	
Young driver:	Male	0.0355 (0.0061)*	—	
	Female	omitted	omitted	
	Age	omitted	omitted	
	17-19	omitted	omitted	
	20-24	-0.0387 (0.0121)*	—	
	25-34	-0.0445 (0.0124)*	—	
	>34	0.0114 (0.0119)	—	
	Experience	<1	omitted	
	1-3	-0.0059 (0.0104)	—	
	>3	0.0762 (0.0121)*	—	
Company year:	First year	omitted	omitted	
	Second year	-0.0771 (0.0122)*	-1.4534 (0.0853)*	
	Third year	-0.0857 (0.0137)*	-2.8459 (0.1191)*	
	Fourth year	-0.1515 (0.0160)*	-3.8089 (0.1343)*	
	Fifth year	-0.4062 (0.0249)*	-3.9525 (0.1369)*	

Note: Standard deviations based on the draws from the posterior distribution in parentheses.

^a σ = Posterior mean of σ across confidence level.

Table 3: Reduced-form and restricted specifications

Table 4: Benchmark structural model

6.7 Table 5: Risk-aversion estimates

Read:

- Table 5 translates the estimated absolute risk-aversion distribution into interpretable dollar lotteries.
- Under the benchmark model, the mean individual is indifferent between a 50-50 gamble of gaining \$100 and losing about \$56.
- The median individual is almost risk neutral for this lottery, willing to lose about \$99.74 against a \$100 gain.
- The large gap between mean and median reflects the right tail of the estimated risk-aversion distribution.
- Relative risk-aversion estimates are very high at the mean but low at the median.

Economic message: The table shows why focusing on a representative individual can be misleading. The average is heavily affected by a minority of very risk-averse consumers, while the median consumer is close to risk neutral for small stakes.

6.8 Table 6: Robustness

Read:

- Table 6 summarizes robustness checks for the utility function, claim-generating process, distribution of risk aversion, information structure, and sample selection.
- A CARA utility specification lowers the estimated mean risk aversion but preserves the qualitative result of substantial heterogeneity.
- Specifications that alter the claim process can change the estimated correlation between risk and risk aversion, but the estimated risk-aversion heterogeneity remains broadly similar.
- Learning and incomplete-information specifications preserve the main qualitative pattern.
- Subsample estimates by company period, referral source, renewal status, and first versus second choice also show similar qualitative conclusions.

Economic message: The robustness table shows which results are stable and which are more model-dependent. The level and heterogeneity of risk aversion are robust; the exact correlation between latent risk and latent risk aversion is more sensitive.

TABLE 5—RISK-AVERSION ESTIMATES

Specification ^a	Absolute risk aversion ^b	Interpretation ^c	Relative risk aversion ^d
Back-of-the-envelope	$1.0 \cdot 10^{-3}$	90.70	14.84
Benchmark model:			
Mean individual	$6.7 \cdot 10^{-3}$	56.05	97.22
25th percentile	$2.3 \cdot 10^{-6}$	99.98	0.03
Median individual	$2.6 \cdot 10^{-3}$	99.74	0.37
75th percentile	$2.9 \cdot 10^{-4}$	97.14	4.27
90th percentile	$2.7 \cdot 10^{-3}$	78.34	39.02
95th percentile	$9.9 \cdot 10^{-3}$	49.37	143.27
CARA utility:			
Mean individual	$3.1 \cdot 10^{-3}$	76.51	44.36
Median individual	$3.4 \cdot 10^{-3}$	99.66	0.50
Learning model:			
Mean individual	$4.2 \cdot 10^{-3}$	68.86	61.40
Median individual	$5.6 \cdot 10^{-6}$	99.95	0.08
Comparable estimates:			
Gertner (1993)	$3.1 \cdot 10^{-4}$	96.99	4.79
Merrick (1995)	$6.6 \cdot 10^{-3}$	99.34	1.02
Holt and Laury (2002) ^e	$3.2 \cdot 10^{-2}$	20.96	865.75
Sydnor (2006)	$2.0 \cdot 10^{-3}$	83.29	53.95

^a The table summarizes the results with respect to the level of risk aversion. “Back-of-the-envelope” refers to the calculation we report in the beginning of Section III. “benchmark model” refers to the results from the benchmark model (Table 4). “CARA utility” refers to a specification of a CARA utility function, and “learning model” refers to a specification in which individuals do not know their risk types perfectly (see Section IIID). The last four rows are the closest comparable results available in the literature.

^b The second column presents the point estimates for the coefficient of absolute risk aversion, converted to $\$/\$^{-1}$ units. For the comparable estimates, this is given by their estimate of a representative CARA utility maximizer. For all other specifications, this is given by computing the unconditional mean and median in the population using the figures we report in the mean and median columns of risk aversion in Table 6 (multiplied by 3.52, the average exchange rate during the period, to convert to US dollars).

^c To interpret the absolute risk aversion estimates (ARA), we translate them into $\{x : u(w) = \frac{1}{2}u(w + 100) + \frac{1}{2}u(w - x)\}$. That is, we report x such that an individual with the estimated ARA is indifferent about participating in a 50–50 lottery of gaining 100 US dollars and losing x US dollars. Note that since our estimate is of absolute risk aversion, the quantity x is independent of w . To be consistent with the specification, we use a quadratic utility function for the back-of-the-envelope, benchmark, and learning models, and use a CARA utility function for the others.

^d The last column attempts to translate the ARA estimates into relative risk aversion. We follow the literature, and do so by multiplying the ARA estimate by average annual income. We use the average annual income (after tax) in Israel in 1995 (51,168 NIS, from the Israeli Census) for all our specifications, and we use average disposable income in the United States in 1997 (\$15,437) for Gertner (1993) and Merrick (1995). For Holt and Laury (2002) and Sydnor (2006) we use a similar figure for 2002 (\$26,974).

^e Holt and Laury (2002) do not report a comparable estimate. The estimate we provide above is based on estimating a CARA utility model for the 18 subjects in their experiment who participated in the “>90” treatment, which involved stakes comparable to our setting. For these subjects, we assume a CARA utility function and a lognormal distribution of their coefficient of absolute risk aversion. The table reports the point estimate of the mean from this distribution.

TABLE 6—ROBUSTNESS

Specification ^a	Sample	Claim risk (λ)				Absolute risk aversion (r)				Corr(r, λ)	p
		Obs.	Mean	Median	Std. dev.	Mean	Median	Std. dev.			
Baseline estimates:											
Benchmark model:	All new customers	105,800	0.220	0.217	0.048	$1.9 \cdot 10^{-3}$	$7.3 \cdot 10^{-6}$	0.020	0.207	0.839	
The VNM utility function:											
CARA utility	All new customers	105,800	0.219	0.217	0.048	$8.7 \cdot 10^{-4b}$	$9.8 \cdot 10^{-6b}$	0.009 ^b	0.201	0.826	
The claim generating process:											
Benchmark model	No multiple claims	103,260	0.182	0.180	0.040	$2.0 \cdot 10^{-3}$	$2.8 \cdot 10^{-3}$	0.018	0.135	0.547	
Benchmark model	Thinner-tail risk distribution	105,800	0.205 ^c	0.171 ^c	0.155 ^c	$1.7 \cdot 10^{-3}$	$1.9 \cdot 10^{-6}$	0.020	−0.076	−0.916	
The distribution of risk aversion:											
Lower-bound procedure	All new customers	105,800	—	—	—	$3.7 \cdot 10^{-4}$	0	0.002	—	—	
Incomplete information about risk:											
Benchmark model	Experienced drivers	82,966	0.214	0.211	0.051	$2.1 \cdot 10^{-3}$	$8.3 \cdot 10^{-6}$	0.021	0.200	0.761	
Benchmark model	Inexperienced drivers	22,834	0.230	0.220	0.073	$3.0 \cdot 10^{-3}$	$1.2 \cdot 10^{-6}$	0.032	0.186	0.572	
Learning model	All new customers	105,800	0.204	0.191	0.084	$1.2 \cdot 10^{-3}$	$1.6 \cdot 10^{-6}$	0.016	0.200	0.772	
Sample Selection:											
Benchmark model	First two years	45,739	0.244	0.235	0.066	$3.1 \cdot 10^{-3}$	$2.6 \cdot 10^{-3}$	0.026	0.225	0.699	
Benchmark model	Last three years	60,061	0.203	0.201	0.043	$1.6 \cdot 10^{-3}$	$3.4 \cdot 10^{-7}$	0.021	0.113	0.611	
Benchmark model	Referred by a friend	26,454	0.213	0.205	0.065	$3.0 \cdot 10^{-3}$	$8.4 \cdot 10^{-7}$	0.031	0.125	0.480	
Benchmark model	Referred by advertising	79,366	0.219	0.216	0.051	$2.1 \cdot 10^{-3}$	$7.6 \cdot 10^{-7}$	0.022	0.212	0.806	
Benchmark model	Non-stayers	48,387	0.226	0.240	0.057	$2.3 \cdot 10^{-3}$	$7.7 \cdot 10^{-7}$	0.026	0.149	0.848	
Benchmark model	Stayers, 1st choice	57,413	0.190	0.182	0.057	$2.9 \cdot 10^{-3}$	$2.9 \cdot 10^{-7}$	0.024	0.152	0.463	
Benchmark model	Stayers, 2nd choice	57,413	0.208	0.200	0.065	$3.0 \cdot 10^{-3}$	$1.6 \cdot 10^{-7}$	0.026	0.211	0.637	

Note: The table presents the key figures from various specifications and subsamples, tracing the order they are presented in Section IIID. Full results (in the format of Table 4) from all these specifications are available in the online Appendix.

^a “Benchmark model” refers to the benchmark specification, estimated on various subsamples (the first row replicates the estimates from Table 4). The other specifications are slightly different, and are all described in more detail in the corresponding parts of Section IIID.

^b The interpretation of r in the CARA model takes a slightly different quantitative meaning when applied to nonfinancial lotteries (such as the approximately

Table 5: Risk-aversion estimates and interpretation

Table 6: Robustness checks

6.9 Table 7: Representativeness

Read:

- Table 7 compares the sample of direct-insurance buyers with the general Israeli adult population and with car owners.
- The sample is more male than the population and more educated than the comparison groups.
- The sample is slightly younger than the car-owner comparison group.
- These differences are consistent with selection into direct insurance.

Economic message: The table cautions against interpreting the estimates as fully representative of all individuals. If direct-insurance customers are less risk averse than the general population, the estimated level of risk aversion may be a lower bound for the broader population.

6.10 Figure 3: Claim distributions

Read:

- Figure 3 plots claim-amount densities for low- and regular-deductible buyers, normalized by the regular deductible.
- About 99% of claims filed by low-deductible buyers are above the regular deductible threshold.
- This means most claims would have been filed under either deductible choice.
- The figure is used to justify abstracting from ex post moral hazard in the main model.

Economic message: The figure supports the paper's focus on claim rates rather than claim amounts. Deductible choice rarely changes whether a claim is filed, because most claims are large relative to both deductible levels.

TABLE 7—REPRESENTATIVENESS

Variable	Sample ^a	Population ^c	Car owners ^d
Age ^a	41.14 (12.37)	42.55 (18.01)	45.11 (14.13)
Female	0.316	0.518	0.367
Family	Single	0.143	0.233
	Married	0.780	0.651
	Divorced	0.057	0.043
	Widower	0.020	0.074
Education	Elementary	0.029	0.329
	High school	0.433	0.384
	Technical	0.100	0.131
	College	0.438	0.155
Emigrant	0.335	0.445	0.447
Obs.	105,800	723,615	255,435

^a For the age variable, the only continuous variable in the table, we provide both the mean and the standard deviation (in parentheses).

^b The figures are derived from Table 1. The family and education variables are renormalized so they add up to one; we ignore those individuals for whom we do not have family status or education level. This is particularly relevant for the education variable, which is not available for about half of the sample; it seems likely that unreported education levels are not random, but associated with lower levels of education. This may help in explaining at least some of the gap in education levels across the columns.

^c This column is based on a random sample of the Israeli population as of 1995. We use only adult population, i.e., individuals who are 18 years of age or older.

^d This column is based on a subsample of the population sample. The data provide information about car ownership only at the household level, not at the individual level. Thus, we define an individual as a car owner if (a) the household owns at least one car and the individual is the head of the household, or (b) the household owns at least two cars and the individual is the spouse of the head of the household.

Table 7: Representativeness of the sample

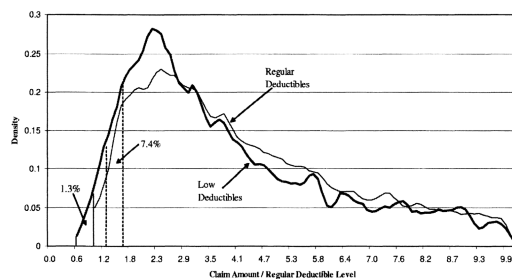


FIGURE 3. CLAIM DISTRIBUTIONS

Notes: This figure plots kernel densities of the claim amounts, estimated separately, depending on the deductible choice. For ease of comparison, we normalize the claim amounts by the level of the regular deductible (i.e., the normalization is invariant to the deductible choice), and truncate the distribution at ten (the truncated part, which includes a fat tail outside of the figure, accounts for about 25 percent of the distribution, and is roughly similar for both deductible choices). The thick line presents the distribution of the claim amounts for individuals who chose a low deductible, while the thin line does the same for those who chose a regular deductible. Clearly, both distributions are truncated from below at the deductible level. The figure shows that the distributions are fairly similar. Assuming that the claim amount distribution is the same, the area below the thicker line between 0.6 and 1 is the fraction of claims that would fall between the two deductible levels, and therefore (absent dynamic incentives) would be filed only if a low deductible were chosen. This area (between the two dotted vertical lines) amounts to 1.3 percent, implying that the potential bias arising from restricting attention to claim rate (and abstracting from the claim distribution) is quite limited. As we discuss in the text, dynamic incentives due to experience rating may increase the costs of filing a claim, shifting the region in which the deductible choice matters to the right; an upper bound to these costs is about 70 percent of the regular deductible, covering an area (between the two dashed vertical lines) that integrates to more than 7 percent. Note, however, that these dynamic incentives are a very conservative upper bound; they apply to less than 15 percent of the individuals, and do not account for the exit option, which significantly reduces these dynamic costs.

Figure 3: Claim distributions and moral-hazard concern

6.11 Figure 4: Counterfactuals - profits

Read:

- Figure 4 plots additional profit per policy from offering a low deductible as a function of the deductible difference.
- The benchmark curve implies that the observed low-deductible option is profitable but not exactly profit-maximizing.
- The observed deductible difference is about 638 NIS and generates about 3.68 NIS of additional profit per policy in the benchmark calculation.
- The profit-maximizing deductible difference in the benchmark is smaller, around 355 NIS.
- Ignoring risk-aversion heterogeneity changes the pricing conclusion dramatically.

Economic message: The figure translates preference estimates into insurance economics. Demand from risk-averse consumers can offset adverse selection, so pricing cannot be understood from risk

heterogeneity alone.

6.12 Figure 5: Counterfactuals - selection

Read:

- Figure 5 decomposes the profit effect into the share of consumers choosing the low deductible and the expected risk of those consumers.
- As the low deductible becomes more attractive, more consumers choose it.
- The expected risk of low-deductible buyers changes with the deductible difference because of selection.
- Under positive correlation between risk and risk aversion, the composition effect differs sharply from the negative-correlation case.

Economic message: The figure shows why multidimensional private information matters. The insurer's marginal buyer is determined by both risk and preferences, so the same price change affects quantity and composition at the same time.

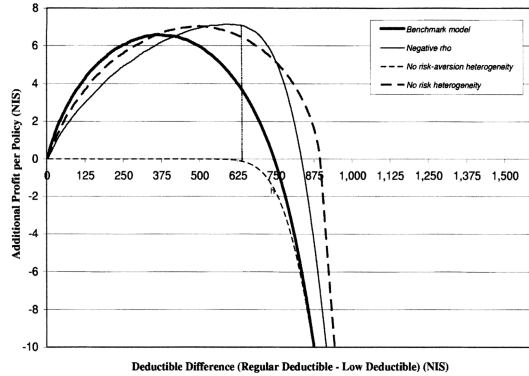


FIGURE 4. COUNTERFACTUALS—PROFITS

Notes: This figure illustrates the results from the counterfactual exercise (see Section IIIF). We plot the additional profits from offering a low deductible as a function of the attractiveness of a low deductible, $\Delta d = d_h - d_l$, fixing its price at the observed level. The thick solid line presents the counterfactual profits as implied by the estimated benchmark model. The other three curves illustrate how the profits change in response to changes in the assumptions: when the correlation between risk and risk aversion is negative (thin solid line), when there is no heterogeneity in risk aversion (dashed line), and when there is no heterogeneity in risk (dotted line). The maxima (argmax) of the four curves, respectively, are 6.59 (355), 5.83 (583), 0, and 7.04 (500). The dotted vertical line represents the observed level of Δd (638), which implies that the additional profits from the observed low deductible are 3.68 NIS per policy.

Figure 4: Pricing counterfactuals and profits

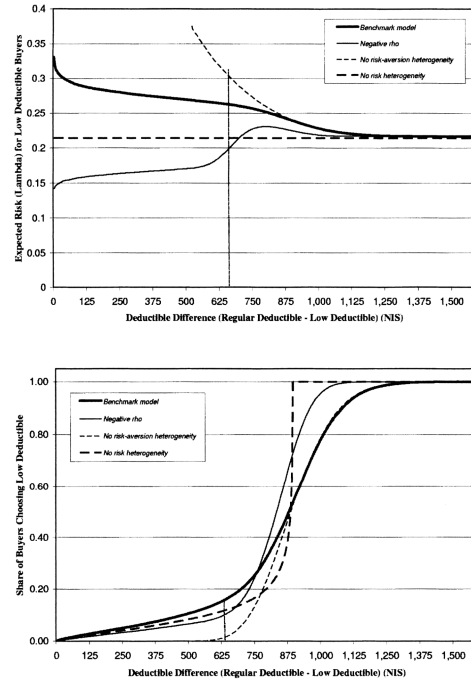


FIGURE 5. COUNTERFACTUALS—SELECTION

Notes: This figure illustrates the results from the counterfactual exercise (see Section IIIF). Parallel to Figure 4, we break down the effects on profits to the share of consumers who choose a low deductible (bottom panel) and to the expected risk of this group (top panel). This is presented by the thick solid line for the estimates of the benchmark model. As in Figure 4, we also present these effects for three additional cases: when the correlation between risk and risk aversion is negative (thin solid line), when there is no heterogeneity in risk aversion (dashed line), and when there is no heterogeneity in risk (dotted line). The dotted vertical lines represent the observed level of Δd (638), for which the share of low deductible is 16 percent and their expected risk is 0.264. This may be compared with the corresponding figures in Table 2A of 17.8 and 0.309, respectively. Note, however, that the figure presents estimated quantities for the average individual in the data, while Table 2A presents the average quantities in the data, so one should not expect the numbers to fit perfectly.

Figure 5: Selection effects in counterfactual pricing

7 Short summary

- The paper estimates risk preferences from auto-insurance deductible choices.

- The data contain 105,800 new policyholders, individual-specific deductible menus, choices, and realized claims.
- The main empirical challenge is that low-deductible buyers may be high risk, highly risk averse, or both.
- The model jointly estimates unobserved claim risk and unobserved absolute risk aversion.
- Claim risk is inferred primarily from realized claims and policy durations.
- Risk aversion is inferred from deductible choices conditional on the inferred risk distribution and menu prices.
- The estimated distribution of risk aversion is highly skewed: the median is close to risk neutral for small stakes, while the mean is much more risk averse.
- Women are more risk averse than men, and risk aversion is U-shaped in age.
- Proxies for income and wealth are often positively related to absolute risk aversion.
- Unobserved risk aversion is more important than unobserved risk for insurance pricing in this setting.
- The counterfactuals show that ignoring preference heterogeneity can lead to misleading conclusions about deductible pricing.

8 Conclusion

8.1 Core conclusion

Cohen and Einav show that actual insurance choices can be used to estimate the distribution of risk preferences. In the Israeli auto-insurance setting, the estimated distribution of absolute risk aversion is highly heterogeneous and strongly right-skewed. The average consumer is meaningfully risk averse over approximately \$100 stakes, but the median consumer is close to risk neutral over such stakes.

8.2 Economic intuition

The intuition is straightforward. A lower deductible is a small insurance product embedded inside the auto policy. It costs more upfront and pays off only if the policyholder has a claim. A high-risk person may buy it even with little risk aversion. A low-risk person who buys it must be relatively risk averse. By combining choices with realized claims, the paper separates these two forces.

8.3 Incremental contribution revisited

The paper moves beyond representative-agent estimates of risk aversion by estimating both observed and unobserved heterogeneity. It also moves beyond reduced-form tests for adverse selection by asking what unobserved dimension drives selection. The important answer is that risk aversion itself is a major source of sorting in insurance demand.

8.4 Implications for insurance markets

The results matter for pricing and contract design. If insurers focus only on claim risk, they miss an important determinant of demand: willingness to pay for certainty. Risk-aversion heterogeneity can make additional coverage profitable even when buyers are adversely selected on risk. This is why multidimensional private information can change optimal insurance menus.

8.5 Limitations

The estimates are local to the auto-insurance deductible context and to relatively small financial stakes. The sample consists of customers who bought from a direct insurer, so it may not represent all consumers. The benchmark assumes expected utility, no moral hazard, and known claim risk. The robustness checks reduce but do not eliminate these concerns.

8.6 Takeaway

The enduring takeaway is that deductible choices reveal substantial heterogeneity in risk preferences. For many consumers, low-deductible demand is not just about being high risk; it is about being highly risk averse. Insurance pricing and welfare analysis should therefore account for both risk heterogeneity and preference heterogeneity.